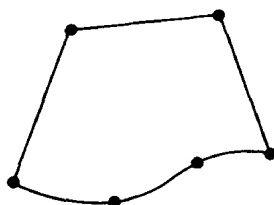


- Exercise 5.** Generalize the flowchart of Exercise 4 so that the 8- through 27-node element may be degenerated to a wedge-shaped element.
- Exercise 6.** Develop a flowchart for a 4- through 16-variable-number-of-nodes quadrilateral element. (The 16-node element is the bicubic Lagrange element of Exercise 2, Sec. 3.6.)
- Exercise 7.** Develop shape functions for a six-node quadrilateral, which exhibits cubic behavior along one edge and linear behavior along the other three (see the accompanying figure).



Standard Element Families

In Fig. 3.7.7, some members of standard two-dimensional element families are shown. (Brick, wedge, and tetrahedral three-dimension analogs may easily be envisioned.) Isoparametric shape functions for all these elements may easily be derived by the techniques described in this section.

We may note from Fig. 3.7.7 that the serendipity elements have nodeless interiors. Beyond cubic level, the serendipity elements may be enhanced by the inclusion of some internal nodes to achieve higher degrees of polynomial completeness. This may be understood by considering the *Pascal triangles* shown in Fig. 3.7.8. As may be seen, for quartic and higher-order serendipity elements, the degree of polynomial completeness is fixed at cubic. The degree of completeness is generally considered to be the most important measure of accuracy of an element. Thus it is seen to be necessary to include functions associated with internal nodes in order to achieve higher-order accurate elements.

3.8 NUMERICAL INTEGRATION; GAUSSIAN QUADRATURE

Let $f: \Omega^e \subset \mathbb{R}^{n_{sd}} \rightarrow \mathbb{R}$ be a given function. We are interested in computing

$$\int_{\Omega^e} f(\mathbf{x}) \, d\Omega \quad (3.8.1)$$

for purposes of constructing element arrays. (f may be thought of as any term in an integrand that we have encountered so far. For example, $f = \mathbf{B}_a^T \mathbf{D} \mathbf{B}_b$, the integrand of the stiffness matrix in heat conduction; see Sec. 2.5.) Throughout we shall assume f is smooth and integrable, so there is no ambiguity as to the meaning of (3.8.1). We

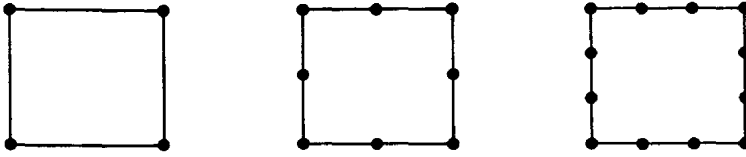
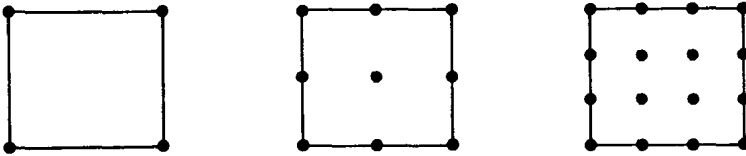
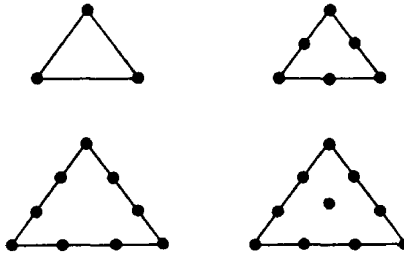
Serendipity family of quadrilateral elementsLagrange family of quadrilateral elementsStandard triangular elements

Figure 3.7.7 Standard two-dimensional element families. Note that the first members of the Lagrange and serendipity families are identical.

will need to make use of the change of variables formula. In each case we will pull back the integral over the element domain in x -space to one over the parent domain (i.e., the biunit n_{sd} -cube). For completeness, we begin by recalling the one-dimensional case (see Sec. 1.15):

$$n_{sd} = 1:$$

$$\int_{\Omega^e} f(x) dx = \int_{-1}^1 f(x(\xi)) x_{,\xi}(\xi) d\xi \quad (3.8.2)$$

The two- and three-dimensional cases are given as follows:

$$n_{sd} = 2:$$

$$\int_{\Omega^e} f(x, y) d\Omega = \int_{-1}^1 \int_{-1}^1 f(x(\xi, \eta), y(\xi, \eta)) j(\xi, \eta) d\xi d\eta \quad (3.8.3)$$

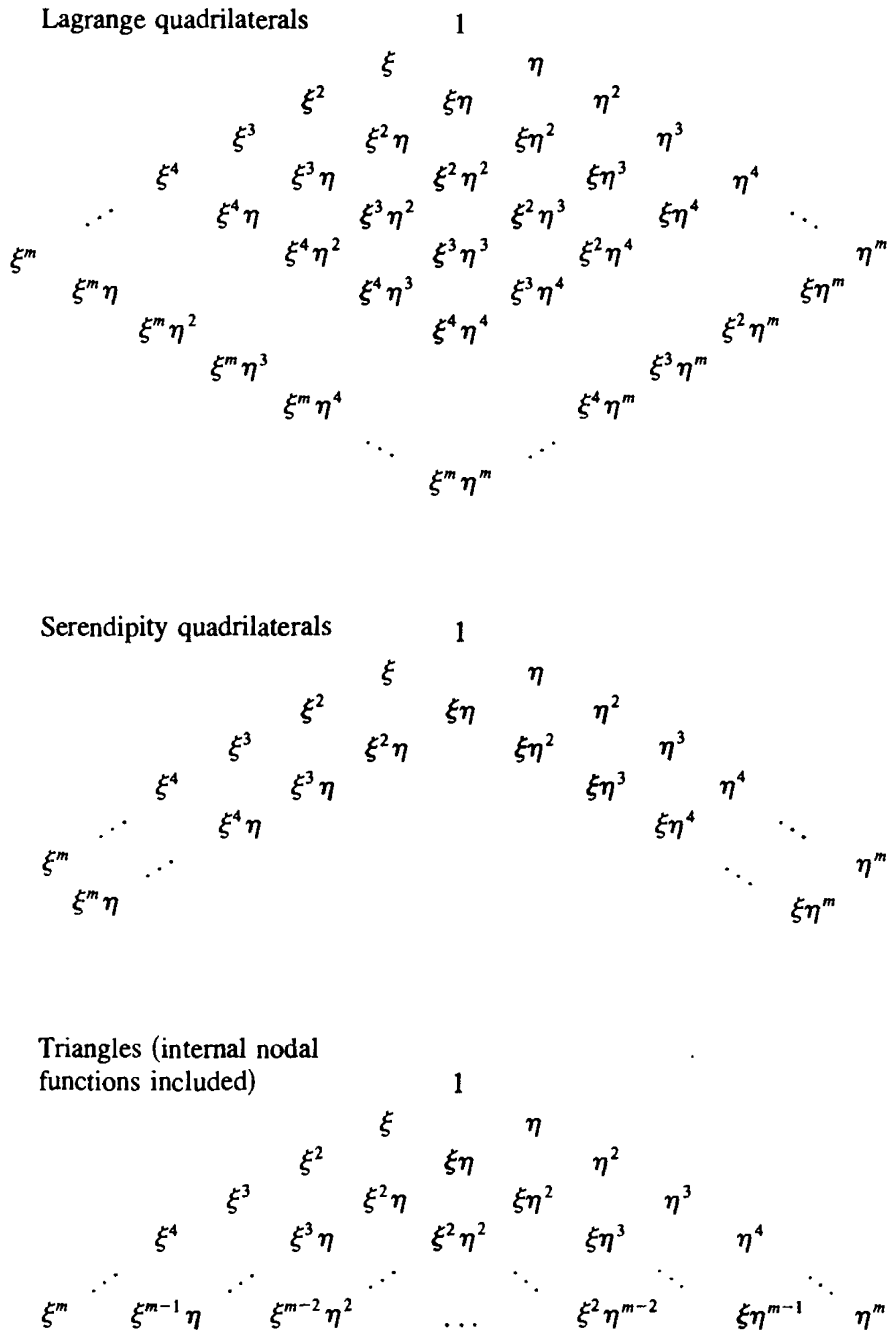


Figure 3.7.8. Pascal triangles for standard two-dimensional element families.

$$n_{sd} = 3:$$

$$\begin{aligned} \int_{\Omega^e} f(x, y, z) d\Omega \\ = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 f(x(\xi, \eta, \zeta), y(\xi, \eta, \zeta), z(\xi, \eta, \zeta)) j(\xi, \eta, \zeta) d\xi d\eta d\zeta \end{aligned}$$

(3.8.4)

Recall that $j = \det(\partial x / \partial \xi)$, the Jacobian determinant.

In each case we have to evaluate an integral of the form:

$$\underbrace{\int_{-1}^1 \cdots \int_{-1}^1}_{n_{sd} \text{ times}} \underbrace{g(\xi, \cdots)}_{n_{sd} \text{ arguments}} \underbrace{d\xi \cdots}_{n_{sd} \text{ differentials}} \quad (3.8.5)$$

Consider the one-dimensional case, in which we wish to evaluate

$$\int_{-1}^1 g(\xi) d\xi \quad (3.8.6)$$

This can be approximately computed by way of a numerical integration (quadrature) formula as follows:

$$\int_{-1}^1 g(\xi) d\xi = \sum_{l=1}^{n_{\text{int}}} g(\tilde{\xi}_l) W_l + R \cong \sum_{l=1}^{n_{\text{int}}} g(\tilde{\xi}_l) W_l \quad (3.8.7)$$

where n_{int} is the number of integration (quadrature) points, $\tilde{\xi}_l$ is the coordinate of the l th integration point, W_l is the “weight” of the l th integration point, and R is the remainder. The reader is probably familiar with some of the classical numerical integration formulas, such as the trapezoidal and Simpson’s rules. We review these now as examples.

Example 1 (Trapezoidal rule)

$$\begin{aligned} n_{\text{int}} &= 2 \\ \tilde{\xi}_1 &= -1 \\ \tilde{\xi}_2 &= 1 \\ W_l &= 1, \quad l = 1, 2 \\ R &= -\frac{2}{3} g_{,\xi\xi}(\bar{\xi}) \end{aligned}$$

$\bar{\xi}$ denotes some point in the interval $[-1, 1]$. The trapezoidal rule is exact for constants

and linear polynomials but only approximate for quadratic and higher-order polynomials. It is said to be "second-order accurate."

Example 2 (Simpson's rule)

$$\begin{aligned}n_{\text{int}} &= 3 \\ \tilde{\xi}_1 &= -1 \\ \tilde{\xi}_2 &= 0 \\ \tilde{\xi}_3 &= 1 \\ W_1 = W_3 &= \frac{1}{3} \\ W_2 &= \frac{4}{3} \\ R &= \frac{-g^{(4)}(\tilde{\xi})}{90}\end{aligned}$$

where $g^{(4)} = g_{\xi\xi\xi\xi}$. Simpson's rule integrates general cubic polynomials exactly and is thus said to be "fourth-order accurate."

Although the above rules are widely used they are inefficient in the sense that there exist integration rules that are just as accurate but involve fewer integration points. *This issue is of great importance in practice since the fewer the integration points, the less the cost.* Considerable savings can be engendered by choosing an appropriate numerical integration rule.

Gaussian Quadrature

In one dimension the Gauss quadrature formulas are optimal. Accuracy of order $2n_{\text{int}}$ is achieved by n_{int} integration points. The locations of the quadrature points and values of associated weights are determined to attain maximum accuracy. The theory is thoroughly discussed in [13]. We give the first three Gaussian rules in the following:

Gaussian quadrature rules.

1. $n_{\text{int}} = 1$

$$\begin{aligned}\tilde{\xi}_1 &= 0 \\ W_1 &= 2 \\ R &= \frac{g_{\xi\xi}(\tilde{\xi})}{3}\end{aligned}$$

2. $n_{\text{int}} = 2$

$$\tilde{\xi}_1 = \frac{-1}{\sqrt{3}}$$

$$\tilde{\xi}_2 = \frac{1}{\sqrt{3}}$$

$$W_1 = W_2 = 1$$

$$R = \frac{g^{(4)}(\tilde{\xi})}{135}$$

$$3. \ n_{\text{int}} = 3$$

$$\tilde{\xi}_1 = -\sqrt{\frac{3}{5}}$$

$$\tilde{\xi}_2 = 0$$

$$\tilde{\xi}_3 = \sqrt{\frac{3}{5}}$$

$$W_1 = W_3 = \frac{5}{9}$$

$$W_2 = \frac{8}{9}$$

$$R = \frac{g^{(6)}(\tilde{\xi})}{15,750}$$

Exercise 1 Verify the accuracy of the preceding Gaussian rules by integrating the monomials 1, ξ , ξ^2 , . . . in turn.

The derivation of Gaussian quadrature formulas is illustrated by the following example.

Example 3 (Derivation of the Gaussian quadrature formula for $n_{\text{int}} = 2$)

We seek a rule involving two integration points that is exact for a general cubic polynomial. Let $g(\xi) = \alpha_0 + \alpha_1 \xi + \alpha_2 \xi^2 + \alpha_3 \xi^3$, where the α 's are arbitrary constants, and assume the integration points are equally weighted and symmetrically spaced (i.e., $W_1 = W_2$ and $\tilde{\xi}_1 = -\tilde{\xi}_2$, respectively). The exact integral of

$$\int_{-1}^1 g(\xi) d\xi = 2\alpha_0 + \frac{2}{3}\alpha_2 \quad (3.8.8)$$

and this is to be equal to

$$\sum_{i=1}^2 g(\tilde{\xi}_i) W_i = 2W_2(\alpha_0 + \alpha_2 \tilde{\xi}_2^2) \quad (3.8.9)$$

This is to hold for arbitrary values of α_0 and α_2 , and it thus follows that $W_2 = 1$ and $\tilde{\xi}_2 = 1/\sqrt{3}$.

Exercise 2. Derive the Gauss quadrature rule for $n_{\text{int}} = 3$. *Hint:* From considerations of symmetry, assume $\tilde{\xi}_1 = -\tilde{\xi}_3$, $\tilde{\xi}_2 = 0$ and $W_1 = W_3$.

General Gaussian Quadrature Rule

The general Gaussian quadrature rule is defined by the following:

$$W_l = \frac{2}{[(1 - \tilde{\xi}_l^2)(P'_{n_{\text{int}}}(\tilde{\xi}_l))^2]}, \quad 1 \leq l \leq n_{\text{int}} \quad (3.8.10)$$

$$R = \frac{2^{2n_{\text{int}}+1}(n_{\text{int}}!)^4}{(2n_{\text{int}}+1)[(2n_{\text{int}}!)]^3} \underbrace{g(\tilde{\xi}_1, \dots, \tilde{\xi}_{n_{\text{int}}})}_{2n_{\text{int}} \text{ times}} \quad (3.8.11)$$

where $\tilde{\xi}_l$ is the l th zero of the Legendre polynomial $P_{n_{\text{int}}}(\xi)$, and $P'_{n_{\text{int}}}$ denotes the derivative of $P_{n_{\text{int}}}$. The Legendre polynomials are defined by

$$P_{n_{\text{int}}}(\xi) = \frac{1}{2^{n_{\text{int}}} n_{\text{int}}!} \frac{d^{n_{\text{int}}}}{d\xi^{n_{\text{int}}}} (\xi^2 - 1)^{n_{\text{int}}} \quad (3.8.12)$$

See [13] for tabulated values of $\tilde{\xi}_l$ and W_l .

Gaussian Quadrature in Several Dimensions

Gaussian rules for integrals in several dimensions are constructed by employing one-dimensional Gaussian rules on each coordinate separately. For example, in two dimensions

$$\begin{aligned} \int_{-1}^1 \int_{-1}^1 g(\xi, \eta) d\xi d\eta &\cong \int_{-1}^1 \left\{ \sum_{l^{(1)}=1}^{n_{\text{int}}^{(1)}} g(\tilde{\xi}_{l^{(1)}}, \eta) W_{l^{(1)}}^{(1)} \right\} d\eta && \text{(Gaussian quadrature rule 1 applied to } \xi\text{-coordinate)} \\ &\cong \sum_{l^{(1)}=1}^{n_{\text{int}}^{(1)}} \sum_{l^{(2)}=1}^{n_{\text{int}}^{(2)}} g(\tilde{\xi}_{l^{(1)}}, \tilde{\eta}_{l^{(2)}}) W_{l^{(1)}}^{(1)} W_{l^{(2)}}^{(2)} && \text{(Gaussian quadrature rule 2 applied to } \eta\text{-coordinate)} \end{aligned}$$

(3.8.13)

Example 4

If the one-point rule is used in each direction (3.8.13) specializes to

$$\int_{-1}^1 \int_{-1}^1 g(\xi, \eta) d\xi d\eta = 4g(0, 0) \quad (3.8.14)$$

Example 5

If the two-point rule is used in each direction, (3.8.13) specializes to

$$\int_{-1}^1 \int_{-1}^1 g(\xi, \eta) d\xi d\eta = g\left(\frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right) + g\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right) + g\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) + g\left(\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \quad (3.8.15)$$

It is frequently preferable to tabulate multidimensional rules in terms of a single index so that in place of (3.8.13) we would have

$$\int_{-1}^1 \int_{-1}^1 g(\xi, \eta) d\xi d\eta \cong \sum_{l=1}^{n_{\text{int}}} g(\tilde{\xi}_l, \tilde{\eta}_l) W_l \quad (3.8.16)$$

Comparison with (3.8.13) indicates that

$$n_{\text{int}} = n_{\text{int}}^{(1)} n_{\text{int}}^{(2)}, \quad 1 \leq l \leq n_{\text{int}} \quad (3.8.17)$$

$$\tilde{\xi}_l = \tilde{\xi}_{l^{(1)}} \quad (3.8.18)$$

$$\tilde{\eta}_l = \tilde{\eta}_{l^{(2)}} \quad (3.8.19)$$

$$W_l = W_{l^{(1)}} W_{l^{(2)}} \quad (3.8.20)$$

where a tabular relationship among the indices must be established (e.g., see Table 3.8.1)

Example 6

If the one-point rule is used in each direction, corresponding to (3.8.17) through (3.8.20), we have

$$n_{\text{int}} = 1, \quad \tilde{\xi}_1 = \tilde{\eta}_1 = 0, \quad W_1 = 2 \cdot 2 = 4 \quad (3.8.21)$$

Thus (3.8.14), (3.8.16), and (3.8.21) are consistent.

Example 7

Consider use of the two-point rule in each direction. We assume the relationship among the indices is as given in Table 3.8.1.

TABLE 3.8.1

l	$l^{(1)}$	$l^{(2)}$
1	1	1
2	2	1
3	2	2
4	1	2

In this case (3.8.17) through (3.8.20) yield

$$n_{\text{int}} = 2 \cdot 2 = 4 \quad (3.8.22)$$

$$\tilde{\xi}_1 = \frac{-1}{\sqrt{3}}, \quad \tilde{\xi}_2 = \frac{1}{\sqrt{3}}, \quad \tilde{\xi}_3 = \frac{1}{\sqrt{3}}, \quad \tilde{\xi}_4 = \frac{-1}{\sqrt{3}} \quad (3.8.23)$$

$$\tilde{\eta}_1 = \frac{-1}{\sqrt{3}}, \quad \tilde{\eta}_2 = \frac{-1}{\sqrt{3}}, \quad \tilde{\eta}_3 = \frac{1}{\sqrt{3}}, \quad \tilde{\eta}_4 = \frac{1}{\sqrt{3}} \quad (3.8.24)$$

$$W_1 = W_2 = W_3 = W_4 = 1 \quad (3.8.25)$$

Clearly (3.8.15), (3.8.16), and (3.8.22) through (3.8.25) are consistent.

Exercise 3. Consider use of the three-point Gauss rule in each direction. Define a relationship between the indices l , $l^{(1)}$, and $l^{(2)}$ by constructing a table similar to Table 3.8.1. Determine $\tilde{\xi}_l$, $\tilde{\eta}_l$, and W_l , $1 \leq l \leq 9$.

Locations of the quadrature points for the one-point, 2×2 , and 3×3 Gaussian rules are illustrated in Fig. 3.8.1.

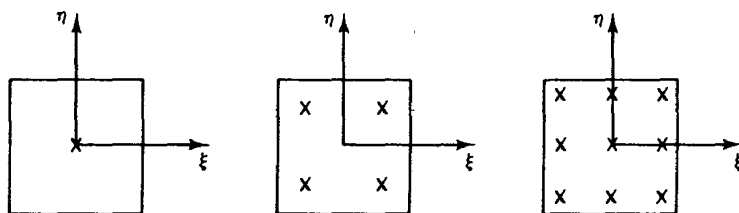


Figure 3.8.1 Locations of Gauss quadrature points for one-point, 2×2 , and 3×3 rules.

Analogous arguments lead to the form of rules in three dimensions, viz.,

$$\begin{aligned} & \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 g(\xi, \eta, \zeta) d\xi d\eta d\zeta \\ & \cong \sum_{l^{(1)}=1}^{n_{\text{int}}^{(1)}} \sum_{l^{(2)}=1}^{n_{\text{int}}^{(2)}} \sum_{l^{(3)}=1}^{n_{\text{int}}^{(3)}} g(\tilde{\xi}_{l^{(1)}}^{(1)}, \tilde{\eta}_{l^{(2)}}^{(2)}, \tilde{\zeta}_{l^{(3)}}^{(3)}) W_{l^{(1)}}^{(1)} W_{l^{(2)}}^{(2)} W_{l^{(3)}}^{(3)} \\ & = \sum_{l=1}^{n_{\text{int}}} g(\tilde{\xi}_l, \tilde{\eta}_l, \tilde{\zeta}_l) W_l \end{aligned} \quad (3.8.26)$$

Exercise 4. Determine $\tilde{\xi}_l$, $\tilde{\eta}_l$, $\tilde{\zeta}_l$, and W_l for the one-point and $2 \times 2 \times 2$ Gaussian rules in three dimensions.

In multidimensional cases the Gaussian quadrature rules are no longer necessarily optimal. For example, in three dimensions the six-point (non-Gaussian) rule given in Table 3.8.2 is fourth-order accurate [14]. The quadrature points are located at the centers of the six faces of the cube. (This means that all monomials of the form

$\xi^i \eta^j \zeta^k$, $0 \leq i, j, k \leq 3$, $i + j + k \leq 3$, are exactly integrated.) The fourth-order Gaussian rule requires eight points ($2 \times 2 \times 2$ rule) and is thus more expensive to use.

TABLE 3.8.2 Fourth-Order Accurate, Six-Point Quadrature Rule in Three Dimensions

l	$\tilde{\xi}_l$	$\tilde{\eta}_l$	$\tilde{\zeta}_l$	w_l
1	1	0	0	4/3
2	-1	0	0	4/3
3	0	1	0	4/3
4	0	-1	0	4/3
5	0	0	1	4/3
6	0	0	-1	4/3

A 14-point rule for three dimensions that is sixth-order accurate is described in [15, 16]. This rule represents a considerable savings when compared with the 27-point ($3 \times 3 \times 3$) Gaussian rule [17].

Unfortunately, a general theory of optimal integration formulas for even such simple shapes as squares and cubes does not yet seem to be known. Despite some measure of inefficiency in multidimensional applications, Gaussian quadrature formulas are still widely used and will be sufficient for our present needs. Recent progress toward the development of improved rules is reported in [18].

Exercise 5. Consider the following four-point rule in two dimensions:

l	$\tilde{\xi}_l$	$\tilde{\eta}_l$	w_l
1	0	+a	1
2	0	-a	1
3	+a	0	1
4	-a	0	1

Determine a such that fourth-order accuracy is attained.

3.9 DERIVATIVES OF SHAPE FUNCTIONS AND SHAPE FUNCTION SUBROUTINES

We need to calculate explicitly the derivatives of the shape functions to construct the element stiffness matrices. For simplicity let us take the case $n_{sd} = 2$. The derivatives